

ABE 715: Computer Programming

Compendium Part 11

Field Deployment & Hardware Guide

Prepared for:
MSc. Irrigation Engineering and Management
MSc. Agricultural and Biosystems Engineering

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Preface to Part 11

While previous modules focused on software logic and data analysis, this final volume addresses the physical reality of engineering: **The Field Environment**.

Agricultural electronics do not live in air-conditioned computer labs; they live in dusty, hot, humid, and off-grid locations. This compendium teaches you how to design robust power systems (Solar/Battery) and protect your code with proper hardware enclosures.

Contents:

- **Power Budgeting:** Calculating solar panel and battery sizes for autonomous operation.
- **Connectivity:** Understanding range, Fresnel zones, and antenna selection for LoRa/WiFi.
- **Hardware Protection:** IP ratings and enclosure design.
- **PCB Design:** Moving from breadboards to printed circuit boards.

Chapter 1

Powering Field Electronics

1.1 The Energy Budget

Before buying a battery, you must calculate your energy consumption.

$$E_{total} = (P_{active} \times t_{active}) + (P_{sleep} \times t_{sleep})$$

Where P is power (Watts) and t is time (Hours).

1.1.1 Example Calculation

An ESP32-based soil sensor node:

- **Active State (WiFi On):** 150mA @ 3.3V = 0.5 Watts. Lasts 10 seconds every hour.
- **Sleep State (Deep Sleep):** 0.01mA @ 3.3V = 0.00003 Watts. Lasts 59 mins 50 secs.

```
def calc_daily_energy(active_current_mA, sleep_current_mA,
    voltage, readings_per_hour):
    # Convert to Amps
    I_active = active_current_mA / 1000
    I_sleep = sleep_current_mA / 1000

    # Time in hours
    t_active_hr = (10 / 3600) * readings_per_hour * 24
    t_sleep_hr = 24 - t_active_hr

    # Energy in Watt-Hours (Wh)
    E_active = (I_active * voltage) * t_active_hr
    E_sleep = (I_sleep * voltage) * t_sleep_hr

    return E_active + E_sleep

energy = calc_daily_energy(150, 0.01, 3.3, 1)
print(f"Daily Consumption: {energy:.4f} Wh")
```

Listing 1.1: Power Budget Calculator

1.2 Solar System Sizing

To run autonomously, the solar panel must generate enough power to recharge the battery and run the load, accounting for cloudy days.

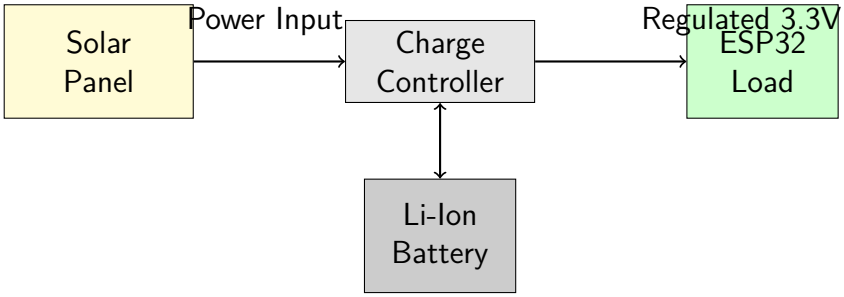


Figure 1.1: Basic Solar Power Circuit Architecture

1.2.1 Sizing Rules of Thumb

- **Battery Capacity (Ah):** Should support the system for 3-5 days without sun (Autonomy Days).
- **Panel Wattage (W):** Should recharge the battery fully in 4-5 peak sun hours.

Chapter 2

Enclosures and Environment

2.1 Ingress Protection (IP) Ratings

Agricultural electronics face rain, dust, and humidity. We use the IP Code to rate enclosures.

Rating	Dust Protection (1st Digit)	Water Protection (2nd Digit)
IP20	Fingers	None (Indoor only)
IP54	Dust Protected	Splashing Water
IP65	Dust Tight	Water Jets (Rain)
IP67	Dust Tight	Immersion (1m depth)

Table 2.1: Common IP Ratings

2.2 Thermal Management

Electronics generate heat. In a sealed IP67 box under the Tanzanian sun, temperatures can exceed 60°C.

- **Vents:** Use Gore-Tex vents that allow air out but stop water entering to prevent condensation.
- **Shields:** Always use a radiation shield (Stevenson Screen) for temperature sensors.

Chapter 3

Wireless Planning

3.1 Technology Selection

Tech	Range	Data Rate	Use Case
WiFi	< 100m	High (Video)	Greenhouse, Home
Bluetooth	< 10m	High	Configuration App
LoRaWAN	1-10 km	Very Low (Bytes)	Remote Soil Sensors
GSM/4G	Unlimited	High	High Value Crops, Video

3.2 The Fresnel Zone

Wireless signals don't travel in a straight line; they travel in a football-shaped zone. If trees or hills block this zone, signal is lost even if you have "Line of Sight".

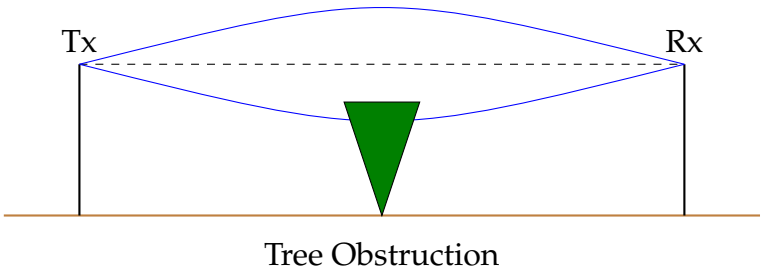


Figure 3.1: Fresnel Zone Clearance

3.3 Antenna Placement

- **Height is King:** Raising an antenna from 1m to 2m can double the range.
- **Polarization:** Ensure both Tx and Rx antennas are oriented the same way (usually vertical).

Chapter 4

PCB Design Basics

4.1 Moving off the Breadboard

Breadboards (white plug-in boards) are for testing. They fail in the field due to vibration and oxidation. For a final project, you should solder components to a **Perfboard** or design a **PCB**.

4.2 PCB Software

- **EasyEDA:** Web-based, very easy for beginners. Good for simple ESP32 shields.
- **KiCad:** Open Source, professional standard.
- **Fritzing:** Good for visualizing wiring, but poor for manufacturing.

4.3 Design Checklist

1. **Power Width:** Make power traces (3.3V, GND) wider to handle current.
2. **Mounting Holes:** Don't forget holes to screw the board into the enclosure.
3. **Labels:** Label all connectors (e.g., "Sensor 1", "Power In") on the silk screen layer.

Chapter 5

Appendix: Hardware Reference

5.1 ESP32 Pinout Guide (DevKit V1)

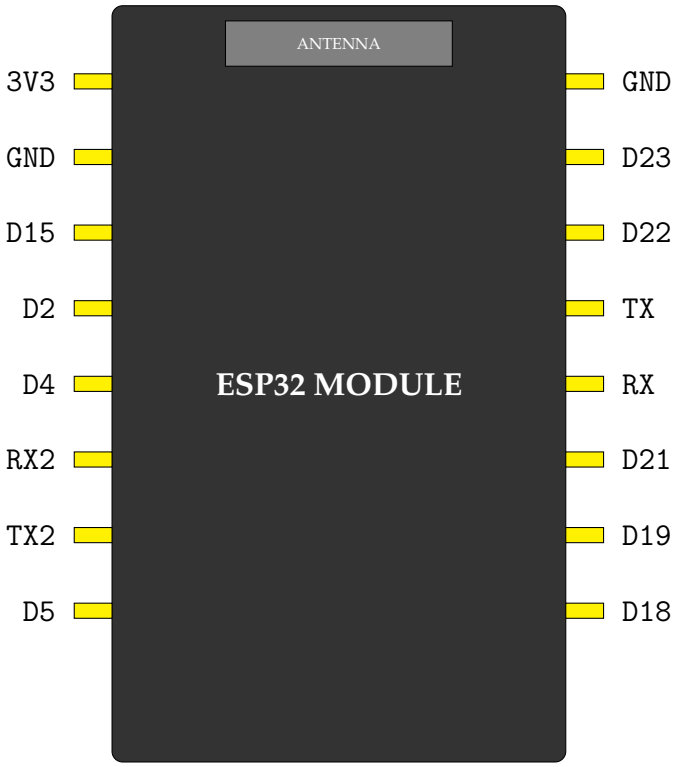


Figure 5.1: Simplified ESP32 Pinout for Agricultural Sensors

5.2 Standard Sensor Wiring Color Code

Wire Color	Function	Voltage
Red	VCC (Power)	3.3V or 5V
Black	GND (Ground)	0V
Yellow/Blue	Data / Signal	0-3.3V
White	Analog Signal	0-3.3V